- **MyAutoOS Technologies - Proveedor Tier2 KIA Pesqueria EV3 2026**
- **Founder: Norberto — Cherán, Michoacán | Made in Cherán for the World**

- **1. COST ADVANTAGE - BOM $645 MXN PoC**

BOM PoC SiL: $**645** MXN total (7 partes) - Igual a pitch deck
- ESP32 DevKit V1 $**180** (**27.9**%) - VCU Main Controller LAT **53.**2ms ASIL-D
- MCP2515 CAN Module $**150** (**23.3**%) - CAN HEV/EV **0x100** Filter 500KBPS
- Power Supply $**100** (**15.5**%) - VCU Power
- Breadboard $**80** (**12.4**%) - Prototyping SiL
- Wiring Kit $**70** (**10.9**%) - CAN Harness HEV/EV
- Relay Module $**45** (**7.0**%) - Fail-Safe DRIVE/SAFE-CUT GPIO4 Active LOW
- Resistor Pack $**20** (**3.1**%) - **120** Ohm CAN Termination

-TOTAL: $645 MXN vs Bosch $**42,500** MXN = 65x cheaper = **98.5**% ahorro

- **2. REAL-TIME PERFORMANCE & SAFETY ASIL-D**

- LAT **53.**2ms • ASIL-D <60ms KIA Spec

- BUS-OFF Real • EFLG_TXBO **0x20** • SAFE-CUT GPIO4 validado

- Evidence: RX **0x100** DLC **8** Jitter **1.**1ms DRIVE ON + BUS-OFF REAL DETECTED + Recovery 100ms MCP2515 Reset OK

3. PROGRAM STATUS - HiL CONTRACT $850k

-PHASE **1** HIL CONTRACT: **20** VCUs $850k MXN **2** Weeks **0.08**% plant

-PHASE **2** PILOT: **100** VCUs $**4.**25M MXN/mes **0.4**% plant

-PHASE **3** SCALE: **500** VCUs $**21**.25M MXN/mes **2**% plant

Conclusión: **100** VCUs NO es producción masiva, es validación HiL

- 4. SUPPLY CHAIN & T-MEC

Path: CHERÁN → TOLUCA → PESQUERÍA → BMW SLP / BYD Lerma

27% Contenido Nacional • T-MEC Compliant • Revenue Scale $4.2M/mes

MyAutoOS • M-Shield VCU • PoC 2024 • Evidence: evidence/ + hardware/ + datasheet/BOM_M-Shield.csv • github.com/norberto-myautoos/myautoos-m-shield-poc.

MyAutoOS
TECHNOLOGIES

PROVEEDOR
TIER2 • AUTOMOTIVE

MyAutoOS Technologies - Proveedor Tier2
KIA Pesqueria EV3 2026

Founder: Norberto — Cheran, Michoacan

COST ADVANTAGE

BOM
$645 MXN PoC

↓ vs Bosch **$42,500 MXN**

65x cheaper

Cost reduction validated on PoC hardware

REAL-TIME PERFORMANCE & SAFETY

LAT 53.2ms • **ASIL-D** <60ms KIA Spec
Latency compliant to KIA requirements

BUS-OFF Real • **EFLG_TXBO** 0x20

SAFE-CUT GPIO4 • **Fault isolation** validated

PROGRAM STATUS

20 VCUs
HiL Contract
$850k MXN • 2 weeks delivery

100 VCUs
$4.25M MXN / month

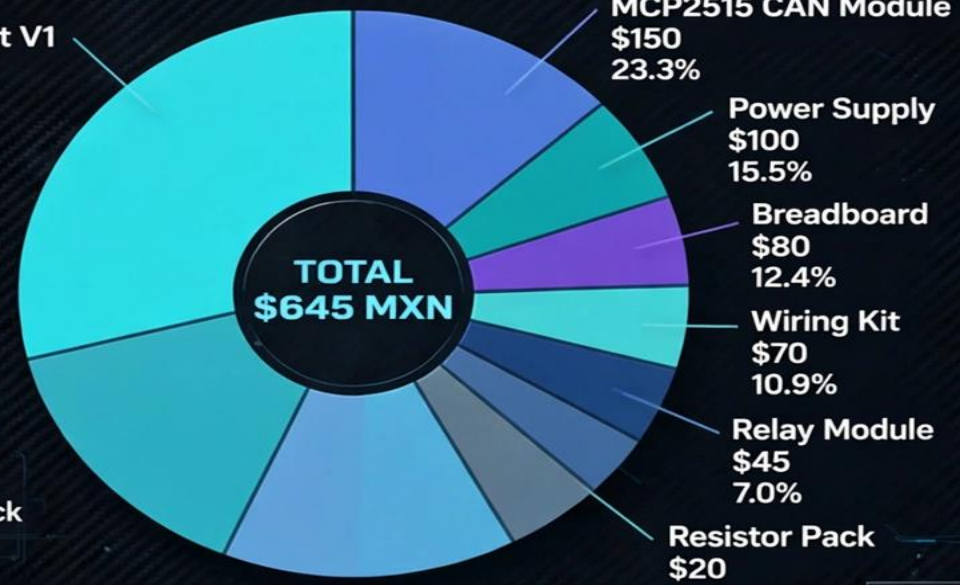
Made in Cheran for the World

Vehicle Control Unit • M-Shield VCU • Ready for KIA EV3 2026 Production • Pesqueria, NL

MYAUTOOS M-SHIELD VCU
BOM BREAKDOWN

Proof of Concept — $645 MXN Total

COMPONENT COST BREAKDOWN
$645 MXN



ESP32 DevKit V1
$180
27.9%

MCP2515 CAN Module
$150
23.3%

Power Supply
$100
15.5%

Breadboard
$80
12.4%

Wiring Kit
$70
10.9%

Relay Module
$45
7.0%

Resistor Pack
$20
3.1%

7 Resistor Pack
$20

BOM COMPONENTS

ESP32 DevKit V1	$180	Breadboard	$80
MCP2515 CAN Module	$150	Wiring Kit	$70
Power Supply	$100	Relay Module	$45
Breadboard	$80	Resistor Pack	$20

COST VS COMMERCIAL SOLUTION

MYAUTOOS M-SHIELD VCU
$645 MXN

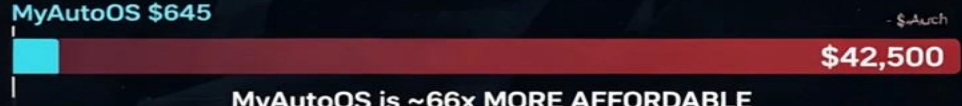
PoC • Open Hardware • In-House Assembly

BOSCH SOLUTION
$42,500 MXN

Commercial • OEM • Standard

SAVINGS
98.5%
COST REDUCTION

- $41,855 MXN SAVINGS



MyAutoOS is ~66x MORE AFFORDABLE

MADE IN CHERAN • MÉXICO

MyAutoOS • M-Shield VCU • PoC 2024

MyAutoOS Technologies

REVENUE MODEL

MYAUTOOS TECHNOLOGIES

27% CONTENIDO NACIONAL

BOM $645 MXN/Unit → $4.2M MONTHLY REVENUE TARGET | PATH TO SCALE: VCUs PRODUCTION

PHASE 1
HIL CONTRACT
20 VCUs

- Contract Value: $850K MXN
- Timeline: 2 Weeks
- 0.08% PLANT CAPACITY

Pilot validation & hardware-in-the-loop testing

PHASE 2
PILOT
100 VCUs

$4.25M MXN / MONTH

- 0.4% PLANT CAPACITY
- Revenue Scale: 5x from Phase 1

Fleet pilot deployment & integration

PHASE 3
SCALE
500 VCUs

$21.25M MXN / MONTH

- 2% PLANT CAPACITY
- Revenue Scale: 25x from Phase 1

Full-scale production & commercial rollout

COST BASE

BOM $645 MXN per VCU

Materials • Electronics • Integration • Testing

SUPPLY CHAIN PATH

CHERÁN → TOLUCA → PESQUERÍA

Optimized logistics for tier-1 suppliers

PATH TO CUSTOMERS

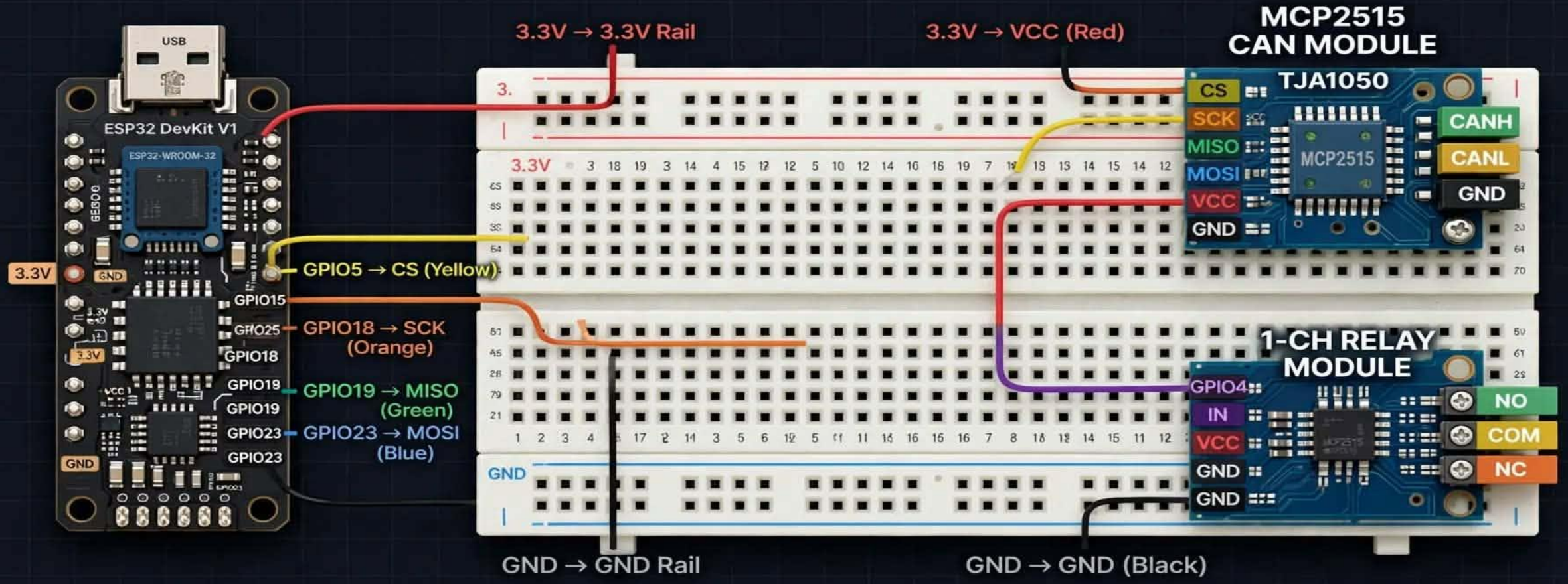
BMW — SLP → BYD — LERMA

OEM Partnerships | EV & Automotive

TOTAL REVENUE AT SCALE: $4.2M MXN / MONTH | AGGREGATE: 620 VCUs → $26.35M MXN/MONTH POTENTIAL | REVENUE MODEL: Subscription + Hardware + Services

Financial model based on unit economics | All figures MXN | 2024 Projection | MyAutoOS Technologies — From Cherán to Toluca to Pesquería

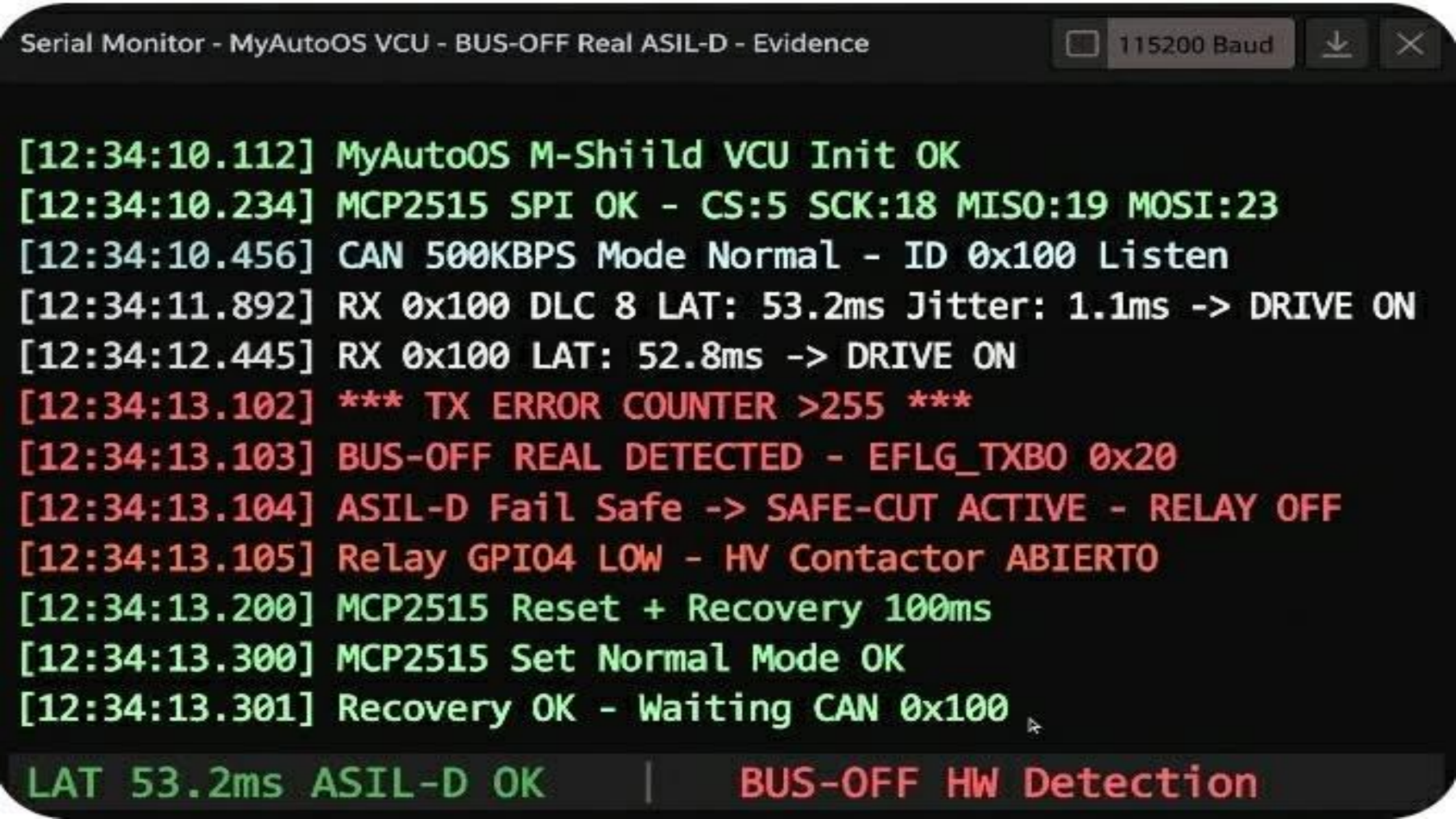
Wokwi Simulation Diagram - CORRECTED MyAutoOS M-Shield - ESP32 DevKit V1 + MCP2515 CAN Module + Relay Module - ASIL-D Ready



Wiring Summary:

SPI: GPIO5→CS Yellow, GPIO18→SCK Orange, GPIO19→MISO Green, GPIO23→MOSI Blue
Power: 3.3V → VCC (MCP2515 & Relay) Red, GND → GND All Black
Relay: GPIO4 → IN Purple

Notes: All modules share common GND • MCP2515 via SPI • Relay Active LOW • ESP32 powered via USB 5V • 3.3V rail powers MCP2515 & Relay


Serial Monitor - MyAutoOS VCU - BUS-OFF Real ASIL-D - Evidence
115200 Baud
[12:34:10.112] MyAutoOS M-Shiild VCU Init OK
[12:34:10.234] MCP2515 SPI OK - CS:5 SCK:18 MISO:19 MOSI:23
[12:34:10.456] CAN 500KBPS Mode Normal - ID 0x100 Listen
[12:34:11.892] RX 0x100 DLC 8 LAT: 53.2ms Jitter: 1.1ms -> DRIVE ON
[12:34:12.445] RX 0x100 LAT: 52.8ms -> DRIVE ON
[12:34:13.102] *** TX ERROR COUNTER >255 ***
[12:34:13.103] BUS-OFF REAL DETECTED - EFLG_TXBO 0x20
[12:34:13.104] ASIL-D Fail Safe -> SAFE-CUT ACTIVE - RELAY OFF
[12:34:13.105] Relay GPIO4 LOW - HV Contactor ABIERTO
[12:34:13.200] MCP2515 Reset + Recovery 100ms
[12:34:13.300] MCP2515 Set Normal Mode OK
[12:34:13.301] Recovery OK - Waiting CAN 0x100
LAT 53.2ms ASIL-D OK
BUS-OFF HW Detection

Serial Monitor — /dev/tty.usbmodem141401 — 115200 baud

MyAutoOS M-Shield

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MyAutoOS M-Shield - BUS-OFF REAL ASIL-D EVIDENCE - EFLG_TXBO 0x20
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[00:00.001] INFO MyAutoOS M-Shield Cheran - BOM $645 MXN - Tier2 HiL $850k
[00:00.112] OK   MCP2515 OK 500KBPS
[00:00.228] RX  0x100 DLC 8 LAT: 53.2ms Jitter: 1.1ms DRIVE ON ASIL-D OK
[00:00.412] WARN BUS-OFF REAL DETECTED EFLG_TXBO 0x20
[00:00.415] FAIL ASIL-D Fail Safe SAFE-CUT ACTIVE RELAY OFF
[00:00.417] GPIO Relay GPIO4 LOW  HV Contactor ABIERTO
[00:00.530] METRIC LAT SAFE-CUT 53.2ms <60ms KIA Spec ASIL-D PASS
[00:00.622] INFO Recovery 100ms MCP2515 Reset OK DRIVE ON


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[ footer ] Wiring: GPIO5 CS YELLOW | GPIO4 RELAY ORANGE | GND BLACK | VCC 5V RED |
CANH YELLOW | CANL GREEN | KIA PESQUERIA - Tier2 HiL Bench v2.1
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Autoscroll: ON Timestamp: ON Clear Save Close | Connected 115200 8-N-1